

Phase 1 Roadmap – AI Code Repository Assistant

Goal: Build the foundation for repository ingestion, code indexing, search, embeddings, and AI Q&A;

Milestone 1: Repository Ingestion

User submits GitHub URL → Validate URL → Create Repository Record in DB → Clone Repository Locally → Track Status → Repository Ready for Scanning. Deliverables: URL submission API, repository schema, cloning service, status tracking.

Milestone 2: Repository Scanner

Scan repository recursively, identify supported files, extract metadata, and store in DB. Metadata: file name, path, extension, size, modified time, language.

Milestone 3: Code Chunking

Parse source code, extract classes, methods, and functions, create chunks, and store them. Store chunk type, name, file path, line range, and source code.

Milestone 4: Keyword Search

Enable search across class names, methods, functions, files, and code content. Deliverables: search API, ranking, and UI.

Milestone 5: Semantic Search & AI Q&A;

Generate embeddings, store vectors, perform semantic search, retrieve relevant chunks, and answer repository questions using AI.

Phase 1 Architecture

GitHub Repository → Repository Ingestion → Repository Scanner → Code Chunking → Metadata Database → Keyword Search → Embeddings → Vector Database → AI Q&A;

Success Criteria

User can submit repositories, repositories are cloned and indexed, code chunks are extracted, keyword search works, vector search works, and AI Q&A; works.